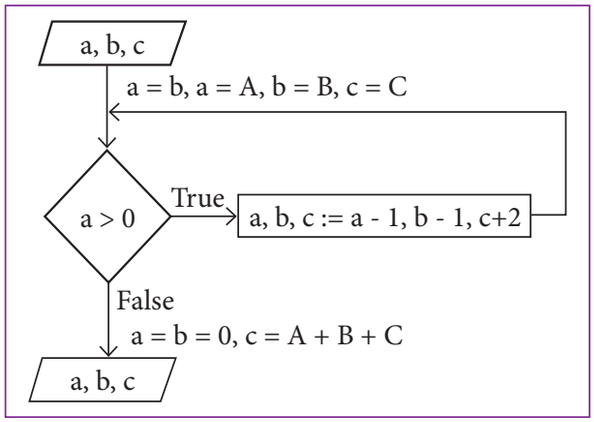
In each meeting, ***a*** and ***b*** each decreases by ***1***, and ***c*** increases by ***2***. The solution can be expressed as an iterative algorithm.

```
monochromatize (a, b,  c)
      -- inputs : a = A, b=B, c=C, a=b
      -- outputs : a = b = 0, c = A+B+C
while a>0
       a, b, c := a-1, b-1, c+2
```

The algorithm is depicted in the flowchart as below



Now let us write this algorithm using function

```
let rec monochromatize a  b  c :=
      if a > 0 then
        a, b, c := a-1, b-1, c+2
      else
      monochromatize a b c
      return a, b, c
```

Points to remember:

- Algorithms are expressed using statements of a programming language
- Subroutines are small sections of code that are used to perform a particular task that can be used repeatedly
- A function is a unit of code that is often defined within a greater code structure
- A function contains a set of code that works on many kinds of inputs and produces a concrete output
- Definitions are distinct syntactic blocks
- Parameters are the variables in a function definition and arguments are the values which are passed to a function definition through the function definition.
- When you write the type annotations the parentheses are mandatory in the function definition
- An interface is a set of action that an object can do
- Interface just defines what an object can do, but won't actually do it
- Implementation carries out the instructions defined in the interface
- Pure functions are functions which will give exact result when the same arguments are passed
- The variables used inside the function may cause side effects though the functions which are not passed with any arguments. In such cases the function is called impure function

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